**Answers to 2026 JC2 Preliminary Examination Paper 2 (H2 Physics)**

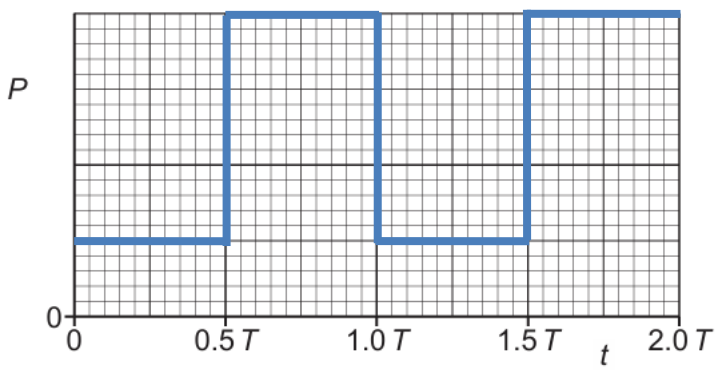
**Suggested Solutions:**

| No. | Solution | Remarks |
|---|---|---|
| **1(a)** | $P = e\sigma AT^4$<br>$e = \frac{P}{\sigma AT^4}$<br>unit of $e = \frac{\text{W}}{(\text{Wm}^{-2}\text{K}^{-4})(\text{m}^2)(\text{K}^4)} = 1$,<br>Hence, *e* has no units. | [1] Correct substitution & presentation |
| **1(b)(i)** | Reduce systematic error: subtract reading from zero error of vernier calipers. | [1] Correct explanation |
| **1(b)(ii)** | $P = e\sigma AT^4$<br>$e = \frac{P}{\sigma(4\pi r^2)T^4}$<br>$e = \frac{6}{(5.67x10^{-8})(4\pi)(2.55x10^{-2})^2(450)^4}$<br>$e = 0.3158$<br><br>$\frac{\Delta e}{e} = \frac{\Delta P}{P} + 2\frac{\Delta d}{d} + 4\frac{\Delta T}{T}$<br>$\frac{\Delta e}{e} = \frac{0.2}{6.0} + 2\frac{0.1}{5.1} + 4\frac{1}{450}$<br>$\frac{\Delta e}{e} = 0.08143$<br>$\Delta e = (0.08143)(0.3158)$<br>$\Delta e = 0.026$<br>$\Delta e = 0.03$<br><br>$e = 0.32 \pm 0.03$ | [B1] correct calculation of *e*<br><br>[M1] correct equation on fractional uncertainty<br><br>[A1] correct calculation of $\Delta e$<br><br>[A1] correct sf for *e* and $\Delta e$ |
| **2(a)** | Gravitational potential at a point as the work done per unit mass by an external force in bringing a small test mass from infinity to that point. | [1] |
| **2(b)** | The gravitational force acting on the satellite by the Earth provides the centripetal force required for the satellite to move in uniform circular motion. | [1] to explain the physics of the situation<br><br>[1] for correct substitutions/equations |

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| | $F_g = F_c$<br>$F_g = ma_c$<br>$\frac{GMm}{r^2} = \frac{mv^2}{r}$<br>$\frac{1}{2}\frac{GMm}{r} = \frac{1}{2}mv^2$<br>$\therefore E_k = \frac{1}{2}\frac{GMm}{r}$ (shown) | |
| **2(c)(i)** | $\frac{1}{2}mv^2 = E_k = \frac{GMm}{2R}$<br>$v = \sqrt{\frac{GM}{R}}$<br>$= \sqrt{\frac{(6.67 \times 10^{-11})(6 \times 10^{24})}{7.2 \times 10^6}}$<br>$= 7455 \text{ m s}^{-1}$<br>$\approx 7500 \text{ m s}^{-1}$ | [1] for correct equation and correct substitution.<br><br>[1] for correct answer |
| **2(c)(ii)** | $v = r\omega = r\left(\frac{2\pi}{T}\right)$<br>$7455 = (7.2 \times 10^6)\left(\frac{2\pi}{T}\right)$<br>$T = 6068 \text{ s} = 0.070 \text{ day} < 1 \text{ day}$<br><br>It is not in geostationary orbit. | [1] for any correct method<br><br>[1] for correct conclusion, to be awarded only if method is correct |
| **2(d)** | As the satellite loses energy and $E_T \propto -\frac{1}{R}$, so its orbital radius decreases.<br><br>Since $E_K \propto \frac{1}{R}$, its kinetic energy and hence speed will increase.<br><br>As speed increases, the work done against resistive forces by the satellite increases and hence converted to more thermal energy, which causes the satellite to burn up. | [1] for relating total energy and orbital radius<br><br>[1] for relating to increasing KE<br>[1] for relating to increasing work done against resistive force |
| **3(a)** | The internal energy $U$ of a system is the sum of a random distribution of microscopic kinetic and potential energies associated with the particles of a system.<br><br>Accept:<br>The internal energy of a system is the sum of the kinetic energy due to the random motion of its molecules, and the potential energy due to intermolecular forces of attraction. | [1] |

<table>
<tr><th>No.</th><th>Solution</th><th>Remarks</th></tr>
<tr><td>3(b)</td><td>compression increase in internal energy $=+2U$<br>thermal energy supplied for compression $=2U - W$<br><br>cooling work done $=0$<br>(at constant volume, $\Delta V = 0 \Rightarrow W = 0$)<br><br>cooling change in internal energy $=-U$<br>thermal energy for cooling $=-U$<br><br><table><tr><td></td><td>work done on gas</td><td>thermal energy supplied to gas</td><td>increase in internal energy of gas</td></tr><tr><td>compression</td><td>$+W$</td><td>$2U - W$</td><td>$+2U$</td></tr><tr><td>cooling</td><td>0</td><td>$-U$</td><td>$-U$</td></tr></table></td><td>[1]<br><br><br><br>[1]</td></tr>
<tr><td>3(c)(i)</td><td>Density $= \dfrac{\text{mass}}{\text{volume}}$<br><br>mass<br>$m = \rho V = (2.70 \times 10^3)(3.61 \times 10^{-3}) = 9.747 \approx 9.75$ kg<br>$V = \dfrac{m}{\rho} = \dfrac{9.747}{2.62 \times 10^3} = 3.7202 \times 10^{-3} = 3.72 \times 10^{-3}$ m$^3$</td><td><br><br><br>[1] for mass or other correct method<br>a<br>[1] for volume</td></tr>
<tr><td>3(c)(ii)</td><td>work done,<br>$W = p\Delta V$<br>$= (1.01 \times 10^5)((3.72 - 3.61) \times 10^{-3})$<br>$= 11.11$<br>$\approx 11.1$ J</td><td>[1] for correct expression and numerical substitution<br>[1] for correct answer</td></tr>
<tr><td>3(c)(iii)</td><td>The volume of the block increases as it expands.<br><br>The block does work against the atmospheric pressure as it pushes the surroundings outward, therefore the work done on the block is negative (work is done by the block on the surroundings).</td><td>[1] for any valid explanation<br><br>[1] for correct conclusion</td></tr>
<tr><td>4(a)(i)</td><td>From the graph, $x = 2.2 \times 10^{-3}$ m<br>$\dfrac{\lambda}{a} = \dfrac{x}{D}$<br>$\dfrac{\lambda}{0.75 \times 10^{-3}} = \dfrac{2.2 \times 10^{-3}}{2.8}$<br>$\lambda = 5.8929 \times 10^{-7}$<br>$\approx 5.9 \times 10^{-7}$ m</td><td>[1] for correct $x$<br><br><br><br>[1] for correct $\lambda$</td></tr>
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| **4(a)(ii)** | It is observed that $2\pi$ rad corresponds to 11 small horizontal squares in Fig 4.2. The phase angle between the two waves arriving at R is 3 small horizontal squares.<br><br>Therefore phase angle<br>$\phi = \frac{3}{11}(2\pi)$<br>$=1.7136$<br>$\approx 1.7$ rad | [1] for correct method<br><br>[1] for correct answer |
| **4(a)(iii)** | Width of each slit is finite and so each slit produces a diffraction pattern, so the intensity varies with angle.<br><br>The interference maxima are modulated by this diffraction envelope, hence they are not all of equal intensity. | [1]<br><br>[1] |
| **4(b)(i)** | $a\sin\theta = n\lambda$<br>$\sin\theta < 1$<br>$\frac{n\lambda}{a} < 1$<br>$n < \frac{a}{\lambda} = \frac{1.60\times10^{-6}}{589.6\times10^{-9}}$ (take the larger wavelength)<br>$n < 2.71$<br>maximum order is 2<br>no. of directions of visible separated light = 8 | [1] for correct method/<br>[1] correct use of wavelength<br>(note: If short wavelength is used, required to find both)<br><br>[1] for correct number of directions |
| **4(b)(ii)** | $a\sin\theta = n\lambda$<br>Angular separation increases with order $n$.<br><br>Greatest separation occurs in the highest order (order 2). | [1] for explanation<br><br>[1] |
| **5(a)** | The root-mean-square (r.m.s.) value of an alternating current is defined as the value of steady direct current that produces heat at the same average rate as the alternating current in a given resistor. | [1] |
| **5(b)(i)** | $I_{\text{r.m.s.}} = \sqrt{\frac{I_0^2(0.5T) + 4I_0^2(0.5T)}{T}}$<br>$=\sqrt{2.5}I_0$<br>$=1.6I_0$ | [1] for correct substitution<br><br>[1] for correct answer |
| **5(b)(ii)** | $\langle P \rangle = I_{\text{r.m.s.}}{}^2 R$<br>$=2.5I_0^2 R$ | [1] for correct answer |

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| **5(b)(iii)** |  | [1] for square wave of period $T$, with $P$ always non-zero<br><br>[1] for horizontal lines, from 0 to 0.5$T$ and from 1.0$T$ to 1.5$T$, all at the same level that the scale indicates to be $I_0^2R$<br><br>[1] for horizontal lines, from 0.5$T$ to 1.0$T$ and from 1.5T to 2.0$T$, at a level that is four times higher than the lower lines |
| **5(c)** | Resistance wires X and Y are connected in series. Hence, same current flows through X and Y.<br><br>Using $I = Anvq$,<br>$n = \frac{I}{Avq} = \frac{I}{\pi r^2 vq}$<br>$\therefore n \propto \frac{1}{r^2 v}$<br><br>$\frac{\text{number density of charge carriers in Y}}{\text{number density of charge carriers in X}}$<br>$\frac{n_Y}{n_X} = \frac{v_X r_X^2}{v_Y r_Y^2}$<br>$= 3 \times \left(\frac{1}{2}\right)^2$<br>$= 0.75$ | [1] for correct expression for $n$ with $r$ and $v$<br><br>[1] for correct substitution<br>[1] for correct answer |
| **6(a)** | Photon is a <u>quantum of electromagnetic radiation with discrete amount of energy which is directly proportional to the frequency</u> of the radiation. | [1] |
| **6(b)** | Threshold frequency refers to the <u>minimum frequency</u> of the incident electromagnetic radiation that will cause photoelectrons to be ejected from the metal surface. | [1] |

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| **6(c)** | Since the interaction between photons and electrons is one-to one, to liberate an electron from the surface, minimum energy is required.<br><br>Since the energy of a photon is proportional to its frequency ($E = hf$), this minimum photon energy requires a minimum frequency, which is the threshold frequency. | [1]<br><br>[1] |
| **6(d)(i)** | $\lambda = \frac{h}{p} = \frac{h}{mv}$<br>$= \frac{6.63 \times 10^{34}}{(9.11 \times 10^{-31})(5.94 \times 10^{6})}$<br>$= 1.23 \times 10^{-10}$ m | [1] sub<br>[1] ans |
| **6d(ii)** | Gain in KE = lost in PE<br>$\frac{1}{2}mv^2 - 0 = eV \rightarrow V = \frac{mv^2}{2e}$<br>$= \frac{(9.11 \times 10^{-31})(5.94 \times 10^{6})^2}{2(1.60 \times 10^{-19})}$<br>$= 100$ V | [1] sub<br>[1] ans |
| **6d(iii)** | To demonstrate electron diffraction, thin crystals are used as natural gratings. The separation between atoms, taken to be the slit separation is of the order of magnitude of $10^{-10}$ m.<br><br>From **(d)(i),** de Broglie wavelength of such electrons is $1.23 \times 10^{-10}$ m, about the same order of magnitude of the slit separation.<br><br>So diffraction effect is well demonstrated. | [1] link the separation between atoms to slit separation<br>[1] expl why diffraction is demonstrated |
| **7(a)(i)** | Displacement $x$ is the projection of radius $R$ on the $x$-axis.<br>$x = R\cos\theta$<br>$= R\cos(\omega t)$ | [1] for explanation relating $\theta$ to $\omega t$ |
| **7(a)(ii)** | Simple harmonic motion of an object is a periodic motion where acceleration is proportional to displacement from an equilibrium position and acceleration is always directed towards the equilibrium position. | [2] |
| **7(a)(iii)** | $x = R\cos(\omega t)$<br>$v = \frac{dx}{dt} = -R\omega\sin(\omega t)$<br>$a = \frac{dv}{dt} = -R\omega^2\cos(\omega t) = -\omega^2\left[R\cos(\omega t)\right] = -\omega^2 x$ | [1] for $v$<br>[1] for $a$ and showing the substitution to s.h.m. |

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| **7(b)(i)** | $A = A_0 e^{-\frac{bt}{2M}}$<br>$\ln A = \ln A_0 - \frac{bt}{2M}$<br>$\ln A = \left(-\frac{b}{2M}\right) t + \ln A_0$<br>This is in the form of $y = mx + c$ | [1] for linearising the equation. |
| **7(b)(ii)** | Gradient is the radio of damping constant $b$ to 2 times of mass $2M$, i.e. gradient $= -\frac{b}{2M}$. | [1] |
| **7(b)(iii)** | At $t = 0$, $\ln(0.12) = -2.1203$<br>At $t = 8.0$, $\ln(0.011) = -4.5099$<br>gradient $= \frac{-2.1203 - (-4.5099)}{0 - 8.0} = -0.29870$<br>$-\frac{b}{2M} = -0.29870$<br>$-\frac{b}{2(2.5)} = -0.29870$<br>$b = 1.4935 \approx 1.5$ kg s$^{-1}$ | [1] for gradient<br>[1] for $b$ |
| **7(b)(iv)** | The graph becomes **steeper**, (or the gradient becomes **more negative**).<br>Greater viscosity means a <u>larger damping constant</u>. | [1]<br>[1] |
| **7(c)(i)** | $F = ma$<br>$kx = m(\omega^2 x)$<br>$k = m\omega^2 = m(2\pi f)^2 = 2.50(2\pi \times 2.1)^2 = 435.25 \approx 440$ N m$^{-1}$ | [1] for correct equation<br>[1] for correct substitution |
| **7(c)(ii)** | Maximum acceleration<br>$a_{max} = \lvert -\omega^2 x_{max} \rvert$<br>$= (2\pi \times 2.1)^2 (0.520)$<br>$= 90.531 \approx$<br>91 m s$^{-2}$ | [1] for correct substitution<br>[1] correct answer |
| **7(d)(i)** | Average power<br>$P_{ave} = \frac{1}{2} b\omega^2 x_0^2$<br>$= \frac{1}{2}(1.5)(2\pi \times 2.1)^2 (0.52)^2$<br>$= 35.307$<br>$\approx 35$ W | [1] correct substitution<br>[1] for correct answer |

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| **7(d)(ii)** | $P = IV = 1.8 \times 24 = 43.2$ W<br>$\text{Efficiency} = \frac{35.307}{43.2} \times 100\%$<br>$= 81.73$<br>$\approx 82\%$ | [1] for P<br><br>[1] |